

DSC 445 — Machine Learning 1

Final Project Report

Predicting U.S. County-Level Presidential Voting Outcomes
Using Demographic and Socioeconomic Features

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Dataset	3,115 U.S. counties — ACS 2020 + MIT Election Lab
Target (Classification)	party_winner (0=Republican, 1=Democrat)
Target (Regression)	dem_vote_share (continuous 0–1)
Best Classification Model	Random Forest — F1: 0.79, AUC: 0.960
Best Regression Model	Random Forest — R ² : 0.737, RMSE: 0.084
Clustering	K-Means k=3 — Silhouette: 0.309
Top Feature	white_pct (MI: 0.112)

1. Introduction

This report documents the complete machine learning pipeline built for DSC 445 to predict U.S. county-level presidential election outcomes using Census demographic and socioeconomic data. The 2020 presidential election was chosen as the target year, combining ACS 5-Year 2020 estimates with MIT Election Lab county-level vote tallies.

The project is structured into five phases: data preprocessing, exploratory data analysis (EDA), classification modeling, regression modeling, and clustering. Each phase builds directly on the previous, forming a cohesive end-to-end ML pipeline.

1.1 Objectives

- Clean and merge raw ACS Census tables with MIT election results into a single county-level dataset
- Identify the demographic and socioeconomic features most predictive of voting outcomes
- Build and compare classification models predicting the binary county winner (`party_winner`)
- Build and compare regression models predicting the continuous Democratic vote share (`dem_vote_share`)
- Apply unsupervised clustering to discover natural county groupings without using election labels

1.2 Data Sources

Source	Description	Features
ACS B01001	Age and sex by county	49 estimate cols
ACS B02001	Racial composition	9 estimate cols
ACS B15003	Educational attainment	24 estimate cols
ACS B19013	Median household income	1 estimate col
ACS B17001	Poverty status	58 estimate cols
ACS B23025	Employment / labor force	7 estimate cols
ACS B25001	Housing units	1 estimate col
MIT Election Lab	County presidential votes 2000-2024	Votes by party

2. Data Preprocessing

The preprocessing pipeline loads 8 raw CSV files, applies consistent cleaning across all ACS tables, filters and reshapes the MIT election data, checks FIPS code alignment between sources, and produces a single merged county-level dataset. Two important bugs were identified and fixed compared to the reference implementation.

2.1 ACS Cleaning Pipeline

Each ACS table required the same set of transformations applied through a shared `clean_acs()` function:

- Drop row 0 — ACS exports include a plain-English description row that is not real data
- Extract `county_fips` — last 5 characters of the `GEO_ID` column (e.g. 0500000US01001 → 01001)
- Keep only estimate columns — filter for columns ending in 'E', exclude `GEO_ID` and `NAME`
- Convert to numeric — `pd.to_numeric(errors='coerce')` handles any leftover text values
- Replace ACS sentinel — -666666666 is the Census Bureau code for suppressed/unavailable data, replaced with NaN
- Zero-pad FIPS — `str.zfill(5)` ensures 1001 becomes 01001 for consistent joining

2.2 Bug Fix 1 — ACS Sentinel Value (-666666666)

The U.S. Census Bureau uses -666666666 as a placeholder when county-level data is suppressed due to small population sizes or privacy concerns. The reference implementation did not handle this value. Without replacement, any mean, correlation, or model computation on affected columns would produce wildly incorrect results. All sentinel values were replaced with NaN before merging.

2.3 Bug Fix 2 — MIT Election Mode Filter

The MIT Election Lab dataset records votes broken down by reporting mode: TOTAL, ABSENTEE, EARLY VOTING, PROVISIONAL, MAIL, and others. The reference approach filtered exclusively for mode == 'TOTAL', which silently dropped 849 counties that report only by individual modes without a TOTAL summary row.

Approach	Counties in MIT Data	Final Merged Counties
Reference (TOTAL only)	2,305	~2,266
Fixed (TOTAL + mode fallback)	3,154	3,115

The fix uses TOTAL rows where available and aggregates all mode rows for counties without a TOTAL record. This recovered all 849 missing counties and aligned the final count with the expected 3,115.

2.4 FIPS Overlap Analysis

Before the final merge, a FIPS overlap check was performed to document exactly which counties would be dropped:

Category	Count	Notes
ACS counties	3,221	Full Census county coverage
MIT counties	3,154	After mode fix
Common (merged)	3,115	Final dataset size
ACS only (dropped)	106	Mostly Alaska boroughs (02xxx FIPS)
MIT only (dropped)	39	Non-standard election reporting units

2.5 Final Dataset

Metric	Value
Total counties	3,115
Total raw features	162
Missing values	1 (B19013_001E — median income, 1 county)
Republican-winning counties	2,576 (82.7%)
Democrat-winning counties	539 (17.3%)
States covered	51 (including DC)

3. Exploratory Data Analysis

3.1 Class Imbalance

The target variable `party_winner` reveals a significant class imbalance that directly impacts modeling strategy:

Class	Counties	Percentage
Republican (0)	2,576	82.7%
Democrat (1)	539	17.3%
Imbalance ratio	4.78x	—

A naive model predicting Republican for every county would achieve 82.7% accuracy while learning nothing useful. All classification models must use `class_weight='balanced'` or equivalent imbalance handling, and evaluation must use F1-score, Precision, Recall, and Confusion Matrix rather than accuracy alone.

3.2 Swing County Analysis

Counties were classified into three competitiveness buckets based on Democratic vote share. This analysis — absent from the reference EDA — provides critical context for model difficulty:

Category	Vote Share Range	Counties	Share
Solid Republican	< 40%	2,204	70.8%
Swing	40% - 60%	673	21.6%
Solid Democrat	> 60%	238	7.6%

The 673 swing counties are the most challenging for prediction and represent genuine political competition. Models should be specifically evaluated on this subset.

3.3 Feature Engineering

Eight interpretable rate and percentage features were engineered from raw ACS counts. Raw counts are misleading for cross-county comparison since larger counties naturally have larger absolute numbers for every demographic variable.

Feature	Formula	Rationale
<code>higher_ed_rate</code>	$(\text{Bachelor} + \text{Master} + \text{Prof} + \text{PhD}) / \text{Pop}$	Education level normalized
<code>poverty_rate</code>	$\text{Below poverty} / \text{Population}$	Poverty normalized for county size
<code>unemployment_rate</code>	$\text{Unemployed} / \text{Labor force}$	True unemployment rate

white_pct	White population / Total population	Racial composition
black_pct	Black population / Total population	Racial composition
asian_pct	Asian population / Total population	Racial composition
log_population	$\log(1 + \text{total population})$	Corrects heavy right skew
housing_density	Housing units / Population	Urbanization proxy

3.4 Mutual Information Feature Ranking

Mutual information was computed to rank features by predictive power for party_winner. Unlike correlation, mutual information captures non-linear relationships and provides a model-agnostic baseline ranking:

Rank	Feature	MI Score	Interpretation
1	white_pct	0.1124	Strongest predictor — negative relationship with Dem vote
2	asian_pct	0.1101	Concentrated in high-Dem urban areas
3	higher_ed_rate	0.1048	Education strongly correlates with Dem voting
4	log_population	0.0868	Larger counties lean Democrat
5	black_pct	0.0730	Higher Black population → more Dem votes
6	median_income	0.0399	Moderate predictor
7	poverty_rate	0.0098	Surprisingly weak standalone predictor
8	unemployment_rate	0.0094	Weak standalone predictor
9	housing_density	0.0076	Weakest predictor

3.5 Correlation Analysis

The correlation heatmap revealed important multicollinearity warnings relevant to linear models:

- white_pct vs black_pct: -0.81 — extremely high negative correlation. Using both in linear models causes instability
- poverty_rate vs median_income: -0.72 — these measure opposite sides of the same economic concept
- poverty_rate vs unemployment_rate: +0.59 — related but not identical
- white_pct vs dem_vote_share: -0.58 — strongest single linear predictor of vote share
- log_population vs dem_vote_share: +0.54 — larger counties lean Democrat

3.6 State-Level Patterns

State	Avg Dem Share	Avg Income	Avg Edu Rate
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DISTRICT OF COLUMBIA	0.921	\$90,842	42.7%
MASSACHUSETTS	0.664	\$80,795	30.2%
HAWAII	0.648	\$80,076	21.6%
NEBRASKA (most Rep.)	0.199	\$55,716	16.2%
OKLAHOMA (0% Dem counties)	0.203	\$49,645	13.3%
WEST VIRGINIA (0% Dem counties)	0.243	\$46,163	12.4%

4. Classification Models

Three classification models were trained to predict the binary county winner (party_winner: 0=Republican, 1=Democrat). All models used class_weight='balanced' or equivalent to handle the 4.78x class imbalance. Features were standardized using StandardScaler. An 80/20 stratified train/test split was used with random_state=42 for reproducibility.

4.1 Models Trained

- Logistic Regression — class_weight='balanced', max_iter=1000
- Random Forest — 200 trees, class_weight='balanced'
- XGBoost — 200 estimators, scale_pos_weight=4.78 (Republican/Democrat ratio)

4.2 Test Set Results

Model	Accuracy	Dem F1	Dem Precision	Dem Recall	ROC AUC
Logistic Regression	0.883	0.727	0.610	0.898	0.952
Random Forest	0.929	0.788	0.820	0.759	0.960
XGBoost	0.917	0.768	0.741	0.796	0.961

4.3 5-Fold Cross-Validation Results

Model	CV F1 Mean	CV F1 Std	Stability
Logistic Regression	0.687	0.022	Most stable
Random Forest	0.749	0.037	Good
XGBoost	0.758	0.037	Highest CV F1

4.4 Feature Importance — Random Forest

Rank	Feature	Importance	Matches MI Ranking
1	higher_ed_rate	0.2295	Yes (MI rank 3)
2	white_pct	0.1979	Yes (MI rank 1)
3	log_population	0.1270	Yes (MI rank 4)
4	asian_pct	0.1155	Yes (MI rank 2)
5	black_pct	0.1019	Yes (MI rank 5)

4.5 Key Findings

- Random Forest is the best classifier — highest test F1 (0.788), highest precision (0.820), highest accuracy (0.929)
- XGBoost edges out Random Forest on cross-validation F1 (0.758 vs 0.749) suggesting it may generalize slightly better
- Logistic Regression achieves surprisingly strong AUC (0.952) showing the features have strong linear separability
- Feature importance from Random Forest closely matches mutual information ranking from EDA — validating the analysis
- All three models achieve $AUC > 0.95$, indicating excellent discrimination ability

5. Regression Models

Four regression models were trained to predict the continuous Democratic vote share (dem_vote_share) ranging from 0.031 to 0.921. This is a harder problem than binary classification as the model must predict the exact vote percentage rather than just the winner.

5.1 Models Trained

- Linear Regression — ordinary least squares baseline
- Ridge Regression — L2 regularization, alpha=1.0
- Lasso Regression — L1 regularization, alpha=0.001
- Random Forest Regressor — 200 trees

5.2 Test Set Results

Model	R ² Score	RMSE	MAE	CV R ² Mean
Linear Regression	0.6688	0.0943	0.0739	0.673
Ridge	0.6688	0.0943	0.0738	0.673
Lasso	0.6708	0.0940	0.0736	0.672
Random Forest	0.7367	0.0841	0.0629	0.743

5.3 Key Findings

- Random Forest is the best regressor — R² 0.737 explains 73.7% of variance in Democratic vote share
- Linear, Ridge, and Lasso are all identical at R² 0.669 — regularization provides no benefit, there is no overfitting with linear models on this dataset
- Random Forest gains +0.068 R² over linear models by capturing non-linear demographic interactions
- RMSE of 0.084 means predictions are on average off by ~8.4 percentage points — reasonable for county-level prediction
- Random Forest CV R² of 0.743 closely matches test R² of 0.737 — no overfitting
- white_pct is the top feature for regression (importance 0.270) confirming it as the dominant predictor across both tasks

6. Clustering Analysis

K-Means clustering was applied to discover natural groupings of counties based purely on demographic features — without using any election labels. The goal was to determine whether demographic similarity alone captures political voting patterns.

6.1 Optimal k Selection

The elbow method and silhouette scores were computed for k=2 to k=10:

k	Silhouette Score	Decision
2	0.3015	Good but too broad
3	0.3094	OPTIMAL — highest silhouette
4	0.1968	Sharp drop — overclustering
5	0.2097	Slight recovery but below k=3
6-10	0.18-0.20	Consistently lower

6.2 Cluster Profiles

Cluster	Profile	Avg Dem Share	Majority Winner	Size
0	Urban/Diverse — high education, high Asian%	0.50	Democrat (slight)	~510
1	Mixed/Swing — high Black%, moderate income	0.42	Republican (slight)	~550
2	Rural/White — high white%, low education	0.26	Republican (strong)	~2,054

6.3 Key Findings

- K-Means with k=3 discovered county groupings that directly map to political lean without ever seeing election data
- Cluster 2 (Rural/White) contains ~66% of all counties and is overwhelmingly Republican (avg 0.26 dem share)
- Cluster 0 (Urban/Diverse) is the only competitive cluster — nearly 50/50 split between parties
- PCA visualization with 2 components explains 63.5% of variance, showing well-separated clusters
- The natural emergence of politically-aligned clusters from demographics alone confirms that race and education are the dominant structural drivers of U.S. county-level voting patterns

7. Comprehensive Model Evaluation

7.1 Classification Summary

Model	Accuracy	Dem F1	AUC	CV F1
Logistic Regression	0.883	0.727	0.952	0.687
Random Forest	0.929	0.788	0.960	0.749
XGBoost	0.917	0.768	0.961	0.758

7.2 Regression Summary

Model	R ²	RMSE	MAE	CV R ²
Linear Regression	0.669	0.094	0.074	0.673
Ridge	0.669	0.094	0.074	0.673
Lasso	0.671	0.094	0.074	0.672
Random Forest	0.737	0.084	0.063	0.743

7.3 Clustering Summary

Metric	Value
Algorithm	K-Means
Optimal k	3
Silhouette Score	0.309
PCA Variance Explained	63.5% (2 components)
Cluster 0 — Urban/Diverse	Avg dem share: 0.50
Cluster 1 — Mixed/Swing	Avg dem share: 0.42
Cluster 2 — Rural/White	Avg dem share: 0.26

8. Conclusions and Recommendations

8.1 Key Findings

- Random Forest is the best model for both classification (F1: 0.788, AUC: 0.960) and regression (R^2 : 0.737) tasks on this dataset
- white_pct, higher_ed_rate, and asian_pct are the top three predictors — consistent across EDA mutual information, classification feature importance, and regression feature importance
- Demographics alone explain 73.7% of the variance in Democratic vote share — a remarkably strong signal
- K-Means clustering discovered politically-aligned county groups without any election labels, confirming that demographic structure drives voting patterns
- The 4.78x class imbalance requires explicit handling — naive accuracy is misleading and F1-score must be the primary metric
- Linear models plateau at $R^2=0.669$ while Random Forest reaches 0.737 — a +0.068 gain from capturing non-linear interactions

8.2 Modeling Recommendations

- Use class_weight='balanced' for all classification models — do not train without imbalance correction
- Avoid using white_pct and black_pct together in linear models — correlation of -0.81 causes multicollinearity
- Avoid using poverty_rate and median_income together — correlation of -0.72
- Use log_population instead of raw population — the raw distribution is heavily right-skewed
- Evaluate specifically on the 673 swing counties (40-60% vote share) — these are the hardest to predict and most valuable
- Consider XGBoost for final deployment — highest CV F1 (0.758) suggests better generalization

8.3 Limitations

- Dataset is limited to 2020 election — trends may not generalize to other election years
- County-level analysis masks within-county variation — urban vs suburban areas within the same county behave differently
- 106 counties dropped due to ACS/MIT coverage gaps — primarily Alaska boroughs
- 9 engineered features capture broad demographic patterns but miss local political history and incumbency effects

8.4 Future Work

- Add historical election data (2012, 2016) as features to capture political momentum

- Incorporate geographic features — proximity to urban centers, state political lean
- Tune hyperparameters with GridSearchCV for XGBoost and Random Forest
- Apply SMOTE oversampling and compare against class_weight approach
- Build a county-level prediction map visualization using FIPS codes